

Elastic Net

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Prerequisites: Linear Regression, Lasso, Ridge.

We define and process $\mathbf{X}, \mathbf{y}, \mathbf{w}$ similar to Linear Regression.

Elastic Net extends Linear Regression, Lasso, Ridge by combining both L_1 and L_2 regularization terms in the loss function. The resulting optimization problem is

$$\underset{\mathbf{w}}{\operatorname{argmin}} \left(|\mathbf{y} - \mathbf{X}\mathbf{w}|^2 + \lambda_1 |\mathbf{w}|_1 + \lambda_2 |\mathbf{w}|_2^2 \right),$$

where

$$\sum_{j=1}^p |w_j|$$

is the L_1 norm of the coefficient vector, and

$$\sum_{j=1}^p w_j^2$$

is the squared L_2 norm. The parameters $\lambda_1 \geq 0$ and $\lambda_2 \geq 0$ control the strengths of the L_1 and L_2 penalties, respectively.

For each coefficient w_j , Elastic Net updates the coefficient using a soft-thresholding step:

$$w_j = \frac{S(\mathbf{x}_j^\top (\mathbf{y} - \mathbf{X}_{-j} \mathbf{w}_{-j}), \lambda_1)}{\mathbf{x}_j^\top \mathbf{x}_j + \lambda_2},$$

where

$$S(\rho, \gamma) = \begin{cases} \rho - \gamma, & \rho > \gamma, \\ 0, & |\rho| \leq \gamma, \\ \rho + \gamma, & \rho < -\gamma. \end{cases}$$

Here, $\mathbf{x}_j$ is the j -th feature column of $\mathbf{X}$, and $\mathbf{X}_{-j}\mathbf{w}_{-j}$ means the prediction using all coefficients except w_j .